DRILL 11

RURAL WATER SUPPLY

NFPA 1001 Level 1 (JPR 4.3.15) / NFPA 1002

DRAFTING FROM A STATIC WATER SOURCE & DRY HYDRANT

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Gear or Work Boots/Gloves (PFD if near water edge)

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Pump Operator & Suction Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to assemble 10 to 20 feet of hard suction hose with a strainer, connect to the pumper's master intake, submerge the strainer into a static water source (pond, river, stream, cistern) or couple to a municipal **dry hydrant**, pull a continuous prime within **30 seconds**, and transition from tank water to draft without cavitation. The foundation of rural firefighting.

📦 TOOLS & APPARATUS SETUP

- Pumper with 2 sections of 6-inch (150mm) hard suction hose and master intake.
- Barrel strainer (with tie-off rope) and floating dock/strainer or dry hydrant adapter.
- Static water source: Station cistern, dry hydrant, local pond, brook, or drafting pit.
- Personal Flotation Devices (PFDs) for any members working within 3 metres (10 feet) of open water.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Water Edge Safety:** Any member working within 10 feet of a steep riverbank, pond edge, or ice shelf **must wear a PFD** (life jacket). Never wear full structural turnout gear onto untested winter ice.

⚠️ **Vacuum Gauge Monitoring:** Monitor compound gauge during priming. If vacuum exceeds 22 inches of mercury (inHg) or fluctuates wildly, cavitation is occurring.

🛑 **THE STOP RULE: Shout "STOP!" immediately if bank soil collapses near the water edge, or if a hard suction hose slips off a coupling.**

PHASE 1: HARD SUCTION COUPLING & STRAINER SELECTION (15 MINUTES) — SLOW DRY RUN

1. **Coupling Hard Suction:**
   - Two firefighters grasp the 10-foot hard suction section.
   - Inspect the thick rubber gasket in the female swivel coupling for dirt or tears.
   - Thread coupling onto the pumper's master intake and torque tight with rubber mallet or spanner wrenches (an air leak as small as a pinhole will prevent pulling a prime!).
2. **Selecting & Rigging the Strainer:**
   - **Barrel Strainer:** For deep water (>4 feet). Requires minimum 2 feet of water above the strainer to prevent surface whirlpools, and 1 foot of clearance from muddy bottom. Tie a rope around the strainer eyelet to control depth.
   - **Floating Strainer:** For shallow streams or ponds (12 to 24 inches deep). Floats on surface and drafts from bottom.
   - **Dry Hydrant Connection:** Couple hard suction directly to the 4.5" or 6" dry hydrant threaded/Storz head.

DRILL 11: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Rural Water Supply • Practical Application

PHASE 2: PULLING THE PRIME (20 MINUTES) — BUDDY COACHING

1. **Pre-Priming Pump Setup:**
   - Shift transmission into **PUMP GEAR** (confirm speedometer reads ~15–20 mph or green "Pump Engaged" light illuminates).
   - Confirm **Tank-to-Pump** valve is **CLOSED** (vital: an open tank valve allows tank water to satisfy the pump, preventing vacuum from reaching the suction hose).
   - Confirm all discharge valves, bleeders, and drain valves are sealed 100% airtight.
2. **Activating the Primer:**
   - Set engine throttle to recommended priming RPM (typically 1,000–1,200 RPM).
   - Pull and hold the **Primer Control Valve**.
   - Observe the **Compound (Intake) Gauge**: Needle should swing counter-clockwise into vacuum, pulling 10 to 20 inches of mercury (inHg).
   - Listen for tone of primer to change from high-pitched whine to deep rumble as water hits the primer vanes.
   - Watch for a steady stream of water discharging from the primer exhaust tube beneath the truck.
   - Once prime is established (steady discharge), release the primer valve smoothly.
   - Master pressure gauge will immediately climb to 50–100 PSI.

PHASE 3: TRANSITION & MAINTAINING CONTINUOUS FLOW (20 MINUTES) — TIMED / STRESS

1. Open a discharge line (or deck gun / crosslay directed back into the water source) slowly.
2. Adjust engine throttle to maintain desired discharge pressure (100–120 PSI).
3. Open the **Pump Cooler / Recirculating Line** slightly to keep pump from overheating if no water is flowing through nozzles.
4. Practice identifying **Cavitation**: Gravelly rattle in the pump casing and needle bouncing on the discharge gauge. Fix immediately by throttling down or reducing flow.

⚠ COMMON PITFALLS & FIREGROUND HAZARDS

- ✗ *Leaving a bleeder valve cracked open:* Even a 1/8-inch drain valve left open admits enough atmospheric air to destroy the pump's vacuum!
- ✗ *Resting strainer in silt/mud:* Sucking mud, leaves, or stones into the pump clogs impellers and burns out shaft seals. Always suspend the strainer using a tie-off rope.
- ✗ *Forgetting winter ice buildup:* In NB winters, dry hydrants can freeze internally from ground frost. Always check dry hydrant barrel with a weighted plumb line before connecting.

🗣 5-MINUTE TAILGATE DEBRIEF

1. ***"How many seconds did it take to achieve prime?", "Did we hear any air hissing around the hard suction coupling gaskets?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Pump drained: Open master pump drain to clear all raw pond water.
- ☐ Flush pump with clean water from onboard booster tank if brackish/muddy water was drafted.
- ☐ Hard suction disconnected, drained, and secured back in brackets.
- ☐ Strainer cleaned of weeds/debris.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS